Frozen Same-Information Scheduling in a Native Mechanical Robot Fixture

Research development draft [1]

Abstract

FINAL AUDITED MECHANICAL SIMULATION COMPONENT. We compare four rule schedulers on sixteen declared native IsaacLab/PhysX mechanical simulation episodes, four fixed instances per rule, after four separate calibration episodes. The original mechanical task, three physical-event timing terms, strict sampled bilateral hold and their conjunction are scored separately. These are prescribed robot skills and calibrated scheduling, not a VLA, weight learning or chemical-endpoint experiment. All sixteen satisfy mechanical prerequisites. Primary T/F/U is 8/2/6; UNKNOWN is not a definite physical violation. Calibrated midpoint and exact joint grid each certify four of four timing tests, but strict hold and primary-plus-hold are zero of sixteen. These paired results establish no exact-grid or learned-policy advantage.

Table 1. All sixteen declared tests. Each method denominator is four. Primary, mechanical task, strict sampled hold and primary+hold are different predicates. All entries use the complete frozen roster.

Rule	T/F/U	Mech.	Hold	Joint+H
Nominal	0/2/2	4/4	0/4	0/4
FIFO prior	0/0/4	4/4	0/4	0/4
Cal. midpoint	4/0/0	4/4	0/4	0/4
Exact joint	4/0/0	4/4	0/4	0/4

1. Frozen experiment and causal scope

The fixed mechanical component uses the original bottle-operation skill with a prospectively qualified hand stiffness of 1600, preserved mechanical criteria and frozen ± 25 ms timing contracts. Only position-target hold timing is varied; physical state and time are not overwritten. Four new calibration instances use dimensionless mass/inertia factors .8/1.2 and FIFO delays 0/12 ticks. Their frozen empirical min/max ranges are not certified model support. Four test instances use .9/1.1 dimensionless factors and the same two FIFO conditions: interpolation, not mass extrapolation. Deadline values 5/5.5 s and duration values 18/18.5 s are tied to these conditions, preventing independent mass/deadline/duration attribution. Source `0c3cff79` and calibration `0bfe857c` are bound before test execution.

All methods share public information, the skill and mechanical controller. Nominal prior and public FIFO correction provide uncalibrated comparators; calibrated midpoint and exact joint grid use the same four-instance calibration. No test outcome adjusts the gains, tolerance, priors or scheduling formulas. Native stepping is 240 Hz. FIFO delays position targets, while gravity compensation and guards remain immediate; this is not whole-communication or actual LLM-inference latency. Host execution cost is reported separately. These frozen bottle-task targets do not demonstrate recovery against an independently observed irreversible external chemical event.

2. Independent physical semantics and denominator

Bilateral contact onset requires the declared contact window; confirmation is recorded separately. Release is the onset of the original supported, unloaded, open-jaw, upright, stable eight-sample window, not first unloading. The elapsed contact-to-release term is a timing relation, not proof of continuous holding. For occurrence intervals $[\ell_a, u_a]$ and $[\ell_b, u_b]$, timing error is

$$[\ell_b - u_a - D,\ u_b - \ell_a - D].$$

Only full containment in the frozen window passes. Partial overlap, missing endpoints and refusal cannot become success. Every declared episode stays in the sixteen-test denominator; the four calibration episodes remain separate. Primary requires all three timing terms and the original mechanism. Strict hold checks every sampled point under its original definition; negative post-opening/pre-release points remain. Their conjunction is reported in the same front table.

[1] Mechanical simulation supplement; chemical and learning validation remain open. Correspondence to: Author details pending.

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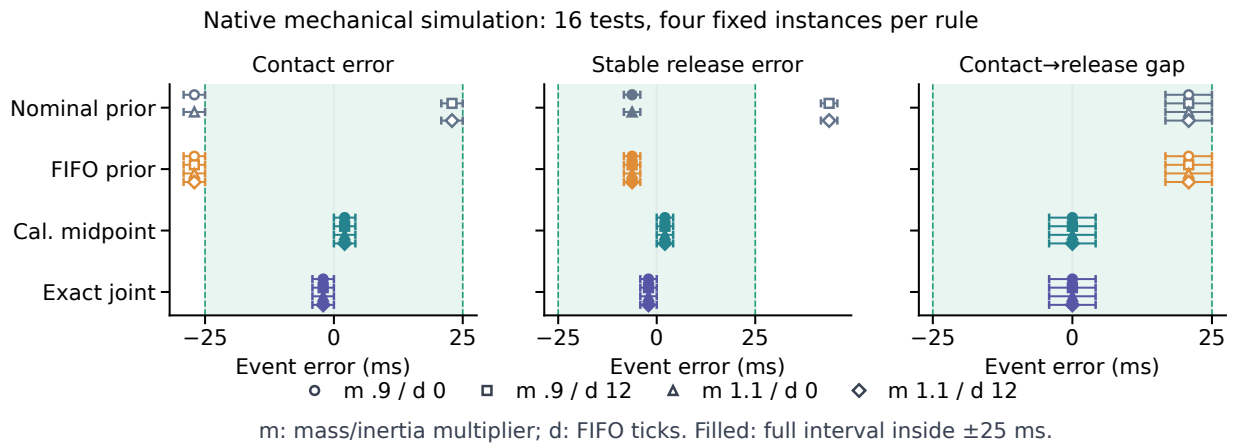


Figure 1. All sixteen tests, three event terms, frozen ±25 ms region. Final markers and error bars come only from the root's complete audited report. A timing pass does not imply strict hold. Mechanical simulation, not RGB, hardware, chemical outcome or VLA evidence.

Table 2. Complete primary T/F/U matrix: four conditions by four rules, all sixteen tests. U is unknown, not a definite violation. m is a dimensionless mass/inertia multiplier; d is FIFO ticks. Full case ids, hold and timing intervals remain in the all-sixteen CSV.

Condition	Nominal	FIFO	Midpoint	Exact
m0.9/d0	U	U	T	T
m0.9/d12	F	U	T	T
m1.1/d0	U	U	T	T
m1.1/d12	F	U	T	T

3. Evidence and attribution limits

Calibrated midpoint and exact joint grid have equal primary counts on these four fixed instances; this matrix does not establish an additional physical benefit from exact optimization. Both prior arms have eight INDETERMINATE contact intervals and eight INDETERMINATE gap intervals combined; only two release intervals are LATE. Their zero certified-primary counts are not evidence of large physical lateness. A prior gap upper bound is about 25.00094 ms against frozen 25 ms: timestamp intervals crossing the boundary remain UNKNOWN; no post hoc tolerance or idealized tick replacement is allowed. Observed contact/release intervals are contained in the empirical calibration box in 16/16 and 16/16 tests. This is not a support or statistical coverage guarantee. Test/calibration task samples: 224,132/53,732; RGBD snapshots: 816/36. Test/calibration allocated GPU-wall upper bounds: 1.171/0.301 h. Snapshots and fixed video frames do not add episodes or native steps.

Four deterministic scene conditions provide paired mechanism evidence, not random-seed statistics, confidence intervals or unseen task-space generalization. Full numeric nine-joint K/D/effort arrays are read at before/after/end snapshots, not continuously per tick. Fresh and cached physical-state preimages are independently checked. Equality at three endpoints cannot establish hidden solver-state or between-snapshot parameter invariance. There are zero new policy weight updates, real-robot episodes and independent chemical endpoints.

An explicit historical correction is retained: all four old calibration traces exported actual K/D/effort arrays at the 1200-step warmup endpoint. The earlier claim of no active readbacks was too broad, including the frozen metadata literal. They still lack the new dedicated after-30 and before/after/end full snapshots; saved comparison flags do not restore numeric arrays. Original sources, scores, failed admissions, budgets and denominators are unchanged. See LEGACY_ACTIVE_READBACK_CORRECTION.json and its Chinese note.

Report binding. Generated numerical content must bind the complete root report, all-sixteen CSV and vector plot, source/calibration hashes and the legacy correction. Report SHA prefix: `39fc19002c08adc7`. This supplement does not establish a new learned algorithm.